LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	27
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Liquid O2 system, O2 charging procedure
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Liquid O2 system, O2 charging procedure
5. Specific Objectives	17.2.1.1 Trainee will be able to understand the need of oxygen systems, types of systems, working principles of components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of O2 charging procedure steps and safety precautions from IB; Instructor supervision of theoretical walkthrough only.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of the aircraft oxygen system?
(b) Name two types of oxygen systems used in aviation.
(c) Why is liquid oxygen preferred over compressed gas for certain aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Need for oxygen systems in aircraft • Types of oxygen systems available • Working principle of Liquid O2 system • Components of Liquid O2 system • O2 charging procedure overview	L & Ds	PPT, IB	15 min
(b)	• Detailed O2 charging procedure steps • Safety precautions during charging • Handling of Liquid Oxygen hazards • Equipment requirements for charging • Post-charging verification checks	L & Ds	PPT, PDF	20 min
(c)	• Common errors in O2 charging • Troubleshooting charging issues • Maintenance requirements post-charge • Documentation and record keeping • Emergency response to leaks	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the working principle of the Liquid O2 system in your own words.
(b) List three critical safety precautions during the O2 charging procedure.
(c) Describe the steps involved in verifying the O2 system after charging.

12. Home Assignment and Follow Up	Prepare a brief report on the maintenance and servicing requirements of the O2 system as covered in the next lesson plan (LP 28).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Maintenance of O2 system, servicing and safety precautions. Safety measures while handling /upkeep of gaseous oxygen system. Quality checks on pressurized gas cylinders.	Refer to the master syllabus reference list	LP 28

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min